

STRESS MONITORING USING EEG AND GSR BIOMARKERS: A REVIEW OF SIGNAL PROCESSING AND AI-BASED APPROACHES

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Abstract

Stress is a common physiological and psychological response that occurs when an individual experiences challenging or demanding situations. Prolonged or excessive stress can affect mental well-being, cognitive performance and overall health. Conventional stress assessment methods often depend on questionnaires and subjective responses, which may not accurately represent the real-time physiological condition of an individual. Therefore, objective monitoring using physiological biomarkers has gained increasing attention. Electroencephalography (EEG) and Galvanic Skin Response (GSR) are two important physiological signals that can provide complementary information about stress. EEG reflects changes in brain electrical activity, while GSR represents changes in skin conductance associated with sympathetic nervous system activity. Recent research has demonstrated the potential of machine learning and deep learning techniques to identify stress-related patterns from these physiological signals. EEG-based stress studies commonly investigate frequency-band power, spectral features, entropy and other time-frequency characteristics, while GSR analysis commonly considers skin conductance level and phasic responses.

This review discusses the physiological basis of stress, EEG and GSR biomarkers, signal preprocessing, feature extraction, machine learning and deep learning approaches, multimodal stress detection and software-based monitoring systems. The review also discusses major challenges including signal noise, individual variability, limited datasets, class imbalance and model generalization. Finally, a software-based framework for stress monitoring using EEG and GSR biomarkers is proposed, in which the processed inputs are analyzed using a trained machine learning model and classified into **low, moderate and high stress levels**.

Keywords: Stress Monitoring, Electroencephalography (EEG), Galvanic Skin Response (GSR), Electrodermal Activity (EDA), Machine Learning, Deep Learning, Physiological Biomarkers, Stress Classification.

1. Introduction

Stress is a natural response of the human body to physical, psychological or environmental demands. A certain level of stress can help an individual respond to challenging situations, but prolonged or excessive stress may negatively affect physical and mental well-being. Stress can influence the autonomic nervous system, cognitive function, emotional state and physiological activity. Because of these effects, reliable stress monitoring has become an important research area in biomedical engineering and healthcare.

Traditional stress assessment commonly depends on self-reported questionnaires and psychological scales. Although these methods are useful for assessing perceived stress, they are subjective and may not continuously represent changes in physiological state. The development of wearable and physiological sensing technologies has therefore created opportunities for more objective stress monitoring. Recent reviews have identified EEG, GSR, ECG, PPG and other physiological measurements as important sources for detecting stress-related responses.

Among these biomarkers, EEG provides information about electrical activity generated by the brain, whereas GSR measures changes in electrical conductance of the skin. These signals represent different aspects of the stress response. EEG can provide information about changes in neural activity, while GSR is closely related to sympathetic nervous system activation and emotional arousal. Combining these signals can therefore provide a more comprehensive representation of stress than using a single physiological signal.

2. Stress Monitoring: Physiological Basis and Biomarkers

2.1 Stress Response and Physiological Changes

When an individual experiences a stressful situation, the body activates physiological mechanisms that help the individual respond to the stressor. The autonomic nervous system plays an important role in this response, particularly through sympathetic nervous system activation.

Stress can result in measurable changes such as increased sweating, changes in heart activity, altered breathing patterns and changes in brain activity. These physiological changes can be measured using biomedical sensors and converted into signals that can be analyzed computationally.

Physiological signals therefore provide an opportunity to detect stress without relying completely on subjective responses. Recent literature emphasizes that multimodal physiological measurements can provide complementary information about stress-related responses.

2.2 EEG as a Stress Biomarker

Electroencephalography (EEG) is a non-invasive technique used to record the electrical activity of the brain using electrodes placed on the scalp. EEG has high temporal resolution, allowing rapid changes in brain activity to be observed.

Stress can influence neural activity and produce changes in different EEG frequency bands. Researchers commonly analyze delta, theta, alpha, beta and gamma components to identify stress-related patterns. Other features such as power spectral density, entropy, statistical characteristics and time-frequency features have also been investigated for stress classification.

Because EEG directly represents neural activity, it can provide information about central nervous system responses that cannot be obtained from peripheral physiological measurements alone.

However, EEG is sensitive to artifacts such as eye movements, muscle activity and environmental electrical interference. Therefore, appropriate preprocessing is necessary before extracting meaningful stress-related features.

2.3 GSR as a Stress Biomarker

Galvanic Skin Response (GSR), also called **Electrodermal Activity (EDA)**, measures changes in the electrical conductance of the skin.

When a person experiences emotional arousal or stress, sympathetic nervous system activity can increase sweat gland activity. Even small changes in sweating can alter skin conductance, which can be measured using GSR electrodes.

GSR is therefore widely used for detecting changes in physiological arousal and stress.

Research on machine learning-based EDA analysis has examined acquisition, preprocessing, feature extraction and classification techniques for stress detection.

GSR has the advantage of being relatively simple to acquire compared with EEG.

However, it is influenced by factors such as temperature, movement, electrode contact and individual differences in sweating response.

2.4 Need for Multimodal EEG and GSR Analysis

EEG and GSR represent different physiological dimensions of the stress response. EEG provides information related to brain activity, while GSR primarily reflects autonomic arousal.

Using both signals can therefore provide complementary information. Multimodal stress detection has become an important research direction because combining multiple physiological sources may improve the robustness of stress classification compared with relying on a single signal. Recent reviews report that multimodal approaches are increasingly being investigated for stress detection, although challenges such as dataset diversity and signal synchronization remain.

The combination of EEG and GSR is particularly relevant for developing an AI-based stress monitoring system because it incorporates both central nervous system information and peripheral autonomic response.

3. EEG and GSR Signal Processing

3.1 EEG Signal Characteristics and Acquisition

EEG signals are low-amplitude electrical signals recorded from the scalp using electrodes positioned according to standardized electrode placement systems. The recorded signal contains information about electrical activity from multiple brain regions. For stress analysis, EEG recordings are generally divided into suitable time segments before feature extraction. Frequency-domain analysis is commonly performed to determine the distribution of signal power across different frequency bands.

Important EEG frequency components include:

- Delta
- Theta
- Alpha
- Beta
- Gamma

Changes in the relative power of these frequency bands can provide useful information for stress classification. Reviews of EEG-based stress detection have identified band power, power spectral density and time-frequency characteristics among the commonly investigated features.

3.2 EEG Preprocessing and Noise Removal

EEG signals are highly sensitive to unwanted interference. Common artifacts include eye movements, muscle activity, electrode movement and power-line interference.

Therefore, preprocessing is performed before feature extraction. Common preprocessing operations include:

- Band-pass filtering
- Notch filtering
- Artifact removal
- Signal normalization
- Segmentation

Filtering can reduce unwanted frequency components, while artifact-removal techniques can reduce interference caused by eye and muscle movements.

Proper preprocessing is important because noisy EEG data can negatively affect machine learning performance. Recent reviews specifically identify noise, uncertainty and variations in EEG acquisition as important challenges in stress detection.

3.3 GSR Signal Processing

GSR signals generally contain slowly varying skin conductance changes and faster responses associated with individual stimuli or events.

The signal can be analyzed using different characteristics such as:

- Skin Conductance Level (SCL)
- Skin Conductance Response (SCR)
- Peak amplitude
- Response frequency
- Rise and recovery characteristics

Preprocessing is required to reduce motion artifacts and other unwanted variations. The cleaned signal can then be divided into suitable time windows for feature extraction.

Machine learning studies on electrodermal activity commonly follow a pipeline involving signal acquisition, preprocessing, feature extraction and classification.

3.4 Feature Extraction from EEG and GSR

Feature extraction converts the raw physiological signals into meaningful numerical parameters that can be provided to a machine learning model.

For EEG, commonly investigated features include:

- Frequency-band power
- Power Spectral Density (PSD)
- Mean and standard deviation
- Entropy
- Hjorth parameters
- Time-frequency characteristics

For GSR, useful features include:

- Mean skin conductance
- Skin conductance level
- Number of responses
- Peak amplitude
- Response duration
- Rise and recovery time

The extracted features from EEG and GSR can either be analyzed independently or combined to form a multimodal feature representation.

4. AI-Based Stress Classification

4.1 Traditional Machine Learning Approaches

Traditional machine learning methods have been widely investigated for physiological stress classification.

Common algorithms include:

- Support Vector Machine (SVM)
- Random Forest
- Decision Tree
- K-Nearest Neighbour (KNN)
- Logistic Regression
- Artificial Neural Network (ANN)

SVM is useful when the number of extracted features is relatively high compared with the available training samples. Random Forest can handle nonlinear relationships and provides an ensemble-based classification approach.

A recent systematic review of workplace stress detection reported frequent use of classical machine learning methods such as Decision Trees, Random Forest and boosting methods, while deep learning models were used in a smaller proportion of reviewed studies.

4.2 Deep Learning Approaches

Deep learning methods can learn complex patterns directly from physiological data or extracted representations.

Common architectures investigated for stress detection include:

- Convolutional Neural Networks (CNN)
- Long Short-Term Memory (LSTM)
- Bidirectional LSTM
- Deep Neural Networks (DNN)

CNN models can learn local patterns, while LSTM-based architectures are useful for sequential physiological signals because they can model temporal relationships.

Recent EEG stress research reviews identify CNN and LSTM among the important deep learning approaches for stress classification.

However, deep learning generally requires larger and well-labelled datasets. Small datasets can increase the risk of overfitting, which remains an important limitation in biomedical stress research.

4.3 Multimodal Feature Fusion

In multimodal stress detection, information from different physiological signals is combined before or during classification.

For EEG and GSR-based monitoring, the extracted EEG features can be combined with GSR features to form a unified feature vector.

The combined representation can then be provided to the trained classification model.

The general concept can be represented as:

EEG features + GSR features → Feature Fusion → ML/DL Model → Stress Level

Multimodal approaches are promising because different biomarkers can capture different components of the physiological response. However, combining modalities also introduces challenges related to synchronization, missing data and differences in signal characteristics.

4.4 Stress-Level Classification

The final classification system can categorize the physiological state into different stress levels.

For the proposed system, three categories are considered:

Low Stress → Moderate Stress → High Stress

The trained model analyzes the EEG and GSR features and produces the corresponding stress category.

The output can then be displayed through a software interface, allowing the user to understand the estimated stress level.

5. Applications of EEG and GSR-Based Stress Monitoring

5.1 Stress-Level Detection

The primary application of EEG and GSR analysis is automated identification of stress-related physiological patterns. Such systems can assist in continuous or repeated monitoring without depending entirely on subjective questionnaires.

5.2 Academic and Workplace Stress Monitoring

Stress monitoring can be applied in academic environments, workplaces and other situations where individuals experience cognitive or emotional pressure.

Wearable and physiological sensing technologies are increasingly investigated for monitoring stress in real-world environments.

5.3 Healthcare Applications

Physiological stress monitoring can potentially support healthcare professionals by providing additional objective information about an individual's stress-related physiological state.

However, such systems should be considered monitoring or decision-support tools rather than standalone diagnostic systems.

5.4 Software-Based Stress Monitoring

AI-based physiological monitoring can be integrated into software applications that allow users to upload or acquire physiological data and obtain an automated classification.

For the proposed system, EEG and GSR inputs are provided to a trained machine learning model, and the resulting stress level is displayed through separate patient and doctor interfaces.

6. Challenges and Future Directions

Several challenges must be addressed before EEG and GSR-based stress monitoring can be reliably used in real-world environments.

Dataset limitations

Many studies use relatively small datasets, which can limit the generalization of trained models. Recent EEG stress reviews have identified limited sample sizes and low statistical power as continuing concerns.

Individual variability

People can respond differently to the same stressful situation. Physiological responses can therefore vary considerably between individuals.

Signal noise and artifacts

EEG is particularly sensitive to eye, muscle and movement artifacts, while GSR can be affected by electrode contact, movement and environmental conditions.

Class imbalance

Stress datasets may contain unequal numbers of low-, moderate- and high-stress samples, which can cause biased classification.

Future directions

Future systems can focus on multimodal sensing, personalized machine learning, explainable AI, larger datasets and real-time monitoring. Recent research is increasingly moving toward multimodal and wearable systems that combine multiple physiological sources.

7. Proposed Methodology

The proposed work focuses on developing a software-based stress monitoring system using EEG and GSR biomarkers.

The overall methodology consists of data acquisition, preprocessing, feature extraction, multimodal analysis, machine learning classification and software implementation.

7.1 Data Acquisition

EEG and GSR datasets are collected from suitable publicly available sources or experimental recordings. The data should contain physiological measurements associated with different stress conditions.

The collected data are organized according to the corresponding stress categories.

7.2 Signal Preprocessing

The EEG signals are processed to reduce unwanted artifacts and interference. Filtering and normalization are applied to improve signal quality.

Similarly, GSR signals are processed to reduce unwanted variations and obtain meaningful skin conductance characteristics.

7.3 Feature Extraction

Relevant features are extracted from both physiological signals.

EEG features include frequency-band power, PSD, statistical features and other time-frequency characteristics.

GSR features include skin conductance level, response amplitude, response frequency and temporal characteristics.

7.4 Multimodal Feature Integration

The extracted EEG and GSR features are combined to create a multimodal representation.

The general processing pipeline is:

EEG Data → Preprocessing → EEG Features

GSR Data → Preprocessing → GSR Features

EEG Features + GSR Features → Feature Fusion

7.5 Machine Learning Classification

The fused feature set is provided to a trained machine learning model.

The model learns the relationship between the physiological features and the corresponding stress categories.

The output consists of:

Low Stress / Moderate Stress / High Stress

Model performance can be evaluated using parameters such as accuracy, precision, recall, F1-score and confusion matrix.

7.6 Software Implementation

The trained model is integrated into a mobile software application developed using Flutter.

The application contains two interfaces:

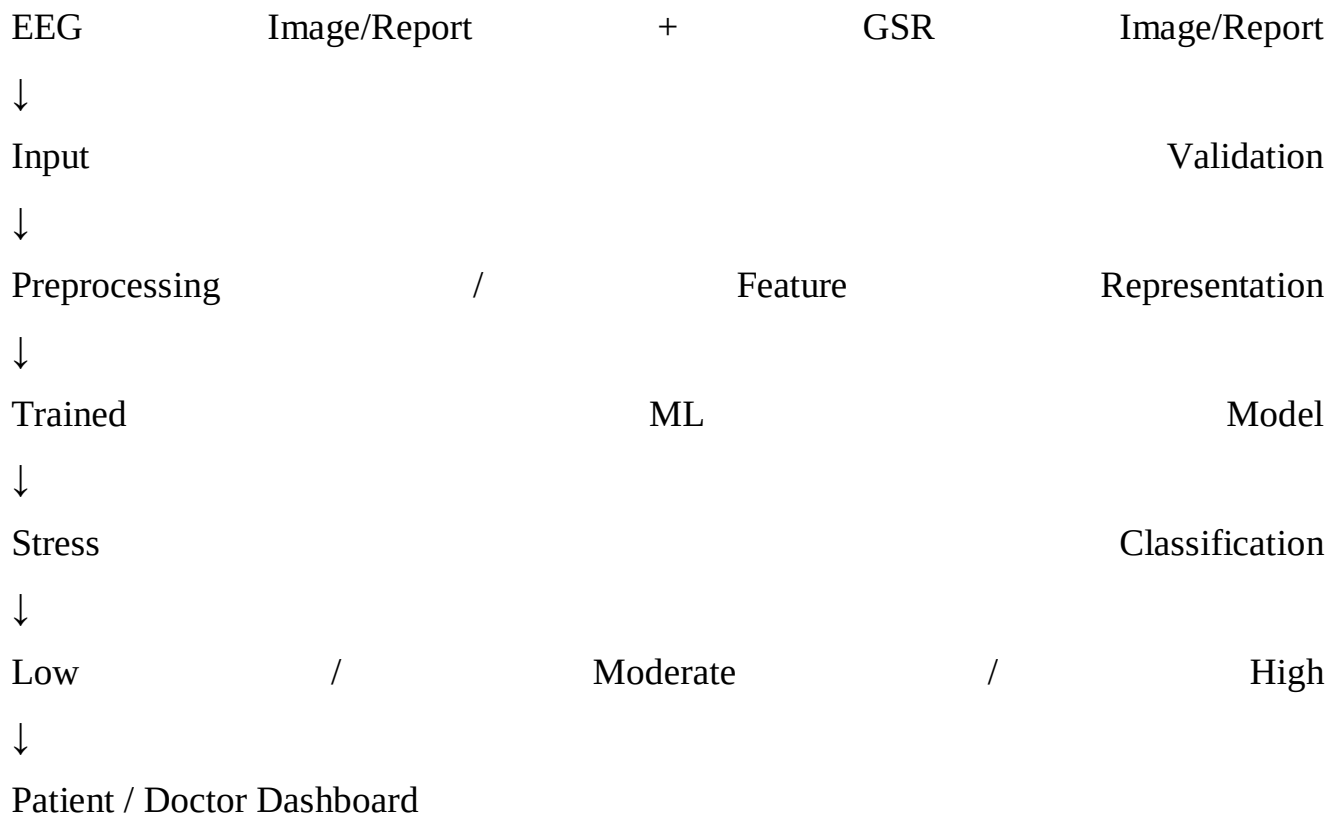
Patient Dashboard

- EEG input
- GSR input
- Stress-level result
- General recommendations

Doctor Dashboard

- EEG input
- GSR input
- Stress-level result
- More detailed stress-related observations

The proposed workflow is:



The application is intended as a research-based stress monitoring and decision-support system, rather than a standalone clinical diagnostic tool.

8. Conclusion and Future Work

EEG and GSR provide complementary physiological information that can be used for objective stress monitoring. EEG provides information about changes in brain electrical activity, while GSR provides information related to sympathetic nervous system activation and physiological arousal. Combining these biomarkers with machine learning can provide a promising approach for automated stress classification.

The review indicates that preprocessing, feature extraction, multimodal fusion and appropriate machine learning algorithms are important components of a reliable stress monitoring system. However, limitations such as small datasets, individual variability, signal artifacts and poor model generalization continue to affect the development of robust systems. Recent systematic reviews similarly emphasize the need for larger, more diverse datasets and stronger validation approaches.

The proposed work aims to address these requirements through a software-based EEG and GSR stress monitoring application capable of classifying stress into low, moderate and high levels. Future development can focus on real-time physiological signal acquisition, personalized models, explainable AI, larger datasets and clinical validation. Such improvements can contribute to more reliable and practical AI-assisted stress monitoring systems.

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